

# ScoreSight - Predicting EPL Points Tally, Top Scorer

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## Abstract -

Football analytics has become an essential component of modern sports management and performance evaluation. The increasing availability of historical football statistics provides opportunities for applying machine learning techniques to predict team performance and individual player achievements. This research presents ScoreSight, an intelligent football analytics system designed to predict English Premier League (EPL) team points tally and identify potential top goal scorers using machine learning algorithms.

The proposed framework utilizes historical EPL datasets containing team and player performance metrics such as goals scored, assists, shots on target, goals per 90 minutes, matches played, and minutes per goal. Data preprocessing techniques including cleaning, normalization, feature selection, and handling missing values are applied to improve prediction accuracy. Multiple machine learning models are evaluated, and the Random Forest Regressor demonstrates superior performance for predicting team points tally, while player performance forecasting is used to identify likely top scorers.

The system is implemented using Python, Scikit-learn, Pandas, NumPy, Flask, HTML, CSS, and JavaScript. A user-friendly web interface enables football enthusiasts, analysts, and researchers to enter performance statistics and obtain predictions in real time. Experimental results indicate that the proposed system achieves high prediction accuracy while maintaining computational efficiency. The framework provides a scalable, interpretable, and data-driven solution for football performance forecasting and sports analytics.

**Index Terms**—Machine Learning, Football Analytics, EPL Prediction, Random Forest, Sports Analytics, Top Scorer Prediction, Data Science, Performance Forecasting

## I. INTRODUCTION

The rapid growth of data analytics and machine learning has transformed the way sports performance is analyzed and predicted. Football, being one of the most popular sports worldwide, generates vast amounts of

statistical data from matches, players, and teams. These data provide valuable insights that can be utilized to forecast future outcomes and support decision-making processes. In recent years, sports analytics has become an important research area where machine learning techniques are applied to analyze historical performance and predict future results.

The English Premier League (EPL) is one of the most competitive football leagues in the world, attracting millions of fans, analysts, and researchers. Predicting team performance and identifying potential top goal scorers are challenging tasks due to the dynamic nature of football matches and the influence of multiple factors such as player form, team strength, match statistics, and historical records. Traditional prediction methods often rely on expert opinions, whereas machine learning approaches can discover hidden patterns within large datasets and generate more accurate predictions.

This research presents **ScoreSight: EPL Team Points Tally and Top Goal Scorer Prediction System**, a machine learning-based application developed to predict the final points tally of EPL teams and identify potential top goal scorers using historical football data. The system utilizes data preprocessing, feature engineering, and machine learning algorithms to analyze team and player statistics. By leveraging historical EPL datasets, the proposed system aims to provide meaningful predictions that can assist football enthusiasts, analysts, and researchers in understanding future league outcomes.

The project incorporates technologies such as Python, Pandas, NumPy, Scikit-learn, Streamlit, HTML, CSS, and JavaScript to create an interactive and user-friendly prediction platform. Data preprocessing techniques are applied to improve dataset quality, while machine

learning models are trained using relevant features extracted from historical football records. The generated predictions are presented through an intuitive web interface, enabling users to explore football analytics efficiently.

The primary objective of this research is to demonstrate the effectiveness of machine learning techniques in sports prediction and football analytics. The proposed system contributes to the growing field of sports data science by providing an automated framework for EPL points tally prediction and top goal scorer forecasting. The results obtained from the system highlight the potential of data-driven approaches in improving prediction accuracy and supporting analytical decision-making in modern football.

## II. RELATED WORK

Sports analytics has emerged as an important research field that combines data science, statistics, and machine learning to analyze and predict sports outcomes. Numerous studies have explored the use of machine learning algorithms for football match prediction, player performance evaluation, and league outcome forecasting. The increasing availability of historical sports datasets has enabled researchers to develop predictive models with improved accuracy and reliability.

Several researchers have applied machine learning techniques such as Linear Regression, Random Forest, Decision Trees, Support Vector Machines, and Neural Networks to predict football match results. These studies demonstrate that historical match statistics, team performance metrics, and player attributes can significantly influence prediction outcomes. Random Forest algorithms, in particular, have gained popularity due to their ability to handle complex datasets and provide accurate predictions with reduced overfitting.

Recent studies have also focused on player performance analysis and goal-scoring prediction. Researchers have utilized player statistics such as goals scored, assists, shots on target, playing time, and historical performance records to identify potential top scorers in football leagues. Machine learning models have shown promising results in forecasting player achievements by analyzing patterns from previous seasons.

Football league prediction systems have been developed to estimate team rankings and season performance based on historical data. These systems commonly use statistical analysis and machine learning techniques to predict points accumulation, league standings, and championship outcomes. Such approaches help analysts, coaches, and

football enthusiasts understand team strengths and potential future performance.

With the advancement of web technologies, several sports analytics platforms have integrated machine learning models into interactive web applications. Frameworks such as Streamlit have enabled researchers to develop user-friendly systems that allow users to input data, visualize statistics, and obtain real-time predictions. These platforms enhance accessibility and provide valuable analytical insights to a broader audience.

Although previous research has achieved significant progress in football prediction, many existing systems focus on either match outcome prediction or player performance analysis individually. Limited studies have combined EPL team points tally prediction and top goal scorer forecasting within a single integrated platform. The proposed ScoreSight: EPL Team Points Tally and Top Goal Scorer Prediction System addresses this gap by utilizing machine learning techniques to predict both team and player performance through an interactive analytics application. The system aims to provide accurate predictions while offering a simple and efficient user interface for football analytics and decision support.

## III. PROPOSED SYSTEM

The proposed system, ScoreSight: EPL Team Points Tally and Top Goal Scorer Prediction System, is designed to leverage machine learning and football analytics to predict team performance and player achievements in the English Premier League (EPL). The system aims to provide accurate forecasts of the final points tally of EPL teams and identify players who are most likely to become the league's top goal scorers. By utilizing historical football statistics and advanced data analysis techniques, the system helps users gain valuable insights into team and player performance.

The development of the proposed system begins with the collection of historical EPL datasets from reliable football statistics sources. These datasets contain comprehensive information related to teams and players, including matches played, goals scored, assists, shots on target, wins, draws, losses, goal differences, and total points earned during previous seasons. The collected data serves as the foundation for building predictive models capable of identifying performance trends and forecasting future outcomes.

Since raw datasets often contain inconsistencies, missing values, duplicate records, and irrelevant information, a detailed data preprocessing phase is performed. During this stage, the data is cleaned and transformed into a structured format suitable for machine learning applications. Missing values are handled appropriately, duplicate entries are removed, and unnecessary features are excluded. The

preprocessing stage improves the overall quality of the dataset and ensures that the machine learning models receive reliable input data for training and prediction.

Feature engineering plays a significant role in improving prediction accuracy. Relevant attributes influencing team success and player scoring ability are carefully selected from the dataset. Statistical indicators such as average goals scored per match, assists, goal conversion rates, team winning percentages, and player performance consistency are analyzed. These features help the machine learning algorithm understand the relationships between historical performance metrics and future outcomes. By selecting meaningful features, the system reduces noise in the dataset and improves predictive efficiency.

The core functionality of the system is powered by machine learning algorithms implemented using Scikit-learn. Historical football data is divided into training and testing datasets, allowing the model to learn patterns from previous seasons and evaluate its performance on unseen data. Random Forest Regression is utilized as the primary prediction algorithm due to its ability to handle large datasets, identify complex relationships among features, and provide robust predictions. The trained model analyzes historical trends and generates predictions regarding the expected points tally of EPL teams as well as potential top goal scorers.

To enhance accessibility and usability, the prediction system is integrated into an interactive web application developed using Streamlit. The user interface is further enhanced using HTML, CSS, and JavaScript to provide a responsive and visually appealing experience. Users can interact with the system through a simple interface, view football statistics, explore historical performance data, and generate predictions with minimal effort. The application presents results in an organized format, making it suitable for football enthusiasts, analysts, students, and researchers.

The system also includes visualization capabilities that help users understand prediction outcomes more effectively. Graphs, charts, and statistical summaries are generated to display team performance trends, player scoring records, and prediction results. Visual representations simplify complex analytical information and enable users to compare historical and predicted performance values. These visualizations support better interpretation of the generated predictions and improve the overall analytical experience.

One of the major advantages of the proposed system is its ability to automate football performance analysis. Traditional forecasting methods often rely on expert judgment and manual analysis, which can be time-consuming and subjective. In contrast, the proposed machine learning-based approach provides data-driven

predictions based on objective statistical evidence. This reduces human bias and improves prediction consistency while enabling faster analysis of large football datasets.

The proposed system is designed to be scalable and adaptable for future enhancements. Additional machine learning algorithms, advanced player performance metrics, real-time data integration, and support for other football leagues can be incorporated into future versions. The flexible architecture of the system allows researchers and developers to extend its functionality according to evolving analytical requirements and technological advancements.

Overall, the ScoreSight system demonstrates how machine learning and sports analytics can be combined to develop an effective football prediction platform. By integrating historical data analysis, feature engineering, predictive modeling, and interactive visualization, the system provides a comprehensive solution for EPL team points tally prediction and top goal scorer forecasting. The proposed system contributes to the growing field of sports data science and highlights the potential of data-driven decision-making in modern football analytics.

### *III. Image Processing and Prediction Workflow*

The Data Processing and Prediction Workflow of the ScoreSight system describes the complete sequence of operations performed to generate EPL team points tally predictions and top goal scorer forecasts. The workflow begins with the collection of historical football datasets containing team and player statistics from previous English Premier League seasons. These datasets serve as the primary source of information for model training and prediction.

After data collection, the datasets undergo a preprocessing stage where missing values, duplicate records, and irrelevant attributes are removed. Data cleaning ensures consistency and improves the quality of information used by the machine learning model. The processed dataset is then transformed into a structured format suitable for analytical operations.

The next phase involves feature engineering, where important performance indicators such as goals scored, assists, matches played, shots on target, wins, draws, losses, goal difference, and points earned are selected. These features help the prediction model understand relationships between historical performance and future outcomes. Feature selection reduces unnecessary complexity and improves model efficiency.

Once feature engineering is completed, the dataset is divided into training and testing sets. The training dataset is used to teach the machine learning algorithm patterns and trends found in historical football statistics. The

testing dataset is used to evaluate model performance and ensure prediction reliability.

The machine learning model, implemented using Random Forest Regression, is then trained using the prepared dataset. During training, the model learns relationships between input features and target variables. After successful training, the model becomes capable of predicting future EPL team points tallies and identifying potential top goal scorers based on historical performance data.

The trained model is integrated into a Streamlit-based application developed using Python, HTML, CSS, and JavaScript. Users can interact with the application through a simple graphical interface. The system accepts user inputs, processes the data, and generates prediction results in real time.

Finally, the prediction results are displayed through tables, charts, and visual reports. Users can analyze team rankings, projected points, player scoring potential, and performance trends. This workflow enables efficient football analytics and provides valuable insights for sports enthusiasts, analysts, and researchers interested in EPL performance prediction.

#### IV. FUNCTIONALMODULES

The ScoreSight system consists of several interconnected functional modules that work together to perform football analytics and prediction tasks. The system begins with the Data Collection module, which gathers historical English Premier League (EPL) datasets containing team and player statistics from reliable football data sources. These datasets include information such as goals scored, assists, matches played, wins, draws, losses, and points accumulated during previous seasons.

After data collection, the Data Preprocessing module prepares the raw datasets for analysis. This module removes missing values, duplicate records, and unnecessary attributes while ensuring data consistency and quality. The cleaned data is then transformed into a structured format suitable for machine learning operations. This step is essential because accurate and reliable predictions depend heavily on the quality of the input data.

The processed data is further handled by the Feature Engineering module, which identifies and selects the most important attributes that influence team performance and player goal-scoring ability. Features such as goals per match, assists, winning percentage, shots on target, goal difference, and total points are extracted and optimized for prediction purposes. Proper feature selection improves model efficiency and enhances prediction accuracy.

The Machine Learning module serves as the core component of the system. Using historical football data,

the Random Forest Regression algorithm is trained to learn relationships between different performance indicators and future outcomes. The trained model analyzes patterns from previous EPL seasons and generates predictions for team points tally and top goal scorer performance. The model is evaluated using testing data to ensure reliability and effectiveness.

To make the prediction system accessible and user-friendly, a web-based interface is developed using Streamlit along with HTML, CSS, and JavaScript. This interface allows users to interact with the system, explore football statistics, and generate predictions through a simple and intuitive environment. Users can easily access analytical information without requiring technical knowledge of machine learning.

The final functional component of the system is the Result Visualization and Analysis module. This module presents prediction outputs in the form of tables, charts, and graphical reports, allowing users to understand team rankings, projected points, and player scoring trends. Through these visual representations, the system provides meaningful insights into football performance and supports data-driven decision-making. Together, these modules form a complete football analytics framework capable of processing historical data, generating predictions, and delivering informative results through an interactive platform.

#### V. SYSTEMARCHITECTURE

The System Architecture of the **ScoreSight: EPL Team Points Tally and Top Goal Scorer Prediction System** is designed to provide a structured workflow for collecting football data, processing information, training machine learning models, and generating predictions through an interactive user interface. The architecture integrates data analytics, machine learning, and web technologies to create a complete football prediction platform.

The system begins with the **Historical EPL Dataset Layer**, which contains team and player statistics collected from previous English Premier League seasons. This dataset includes information such as matches played, wins, draws, losses, goals scored, assists, shots on target, goal difference, and total points. The collected data acts as the primary source for training and evaluating the prediction models.

The dataset is passed to the **Data Preprocessing Layer**, where missing values, duplicate records, and irrelevant attributes are removed. Data cleaning and transformation operations are performed to improve data quality and consistency. This layer ensures that the information used for machine learning is accurate and suitable for analysis.

Proper preprocessing enhances model performance and reduces prediction errors.

After preprocessing, the data enters the Feature Engineering Layer. In this stage, important features influencing team and player performance are selected and optimized. Attributes such as goals per match, assists, winning percentage, goal difference, and player scoring records are extracted. Feature engineering helps the model identify meaningful patterns and relationships within the historical football data.

The processed data is then forwarded to the Machine Learning Layer, where the Random Forest Regression algorithm is implemented using Scikit-learn. The model is trained using historical EPL statistics and learns patterns associated with team success and player scoring performance. Once training is completed, the model generates predictions for team points tally and top goal scorer outcomes. The model is also evaluated using testing datasets to measure prediction accuracy and reliability.

The prediction results are integrated with the Application Layer, which is developed using Streamlit along with HTML, CSS, and JavaScript. This layer provides a user-friendly interface through which users can interact with the system. Users can explore football statistics, initiate predictions, and access analytical insights without requiring technical expertise in machine learning.

Finally, the Visualization and Output Layer presents prediction results through tables, charts, and graphical reports. Users can view projected team standings, estimated points tallies, and top goal scorer predictions in an organized format.

## VI. SYSTEM IMPLEMENTATION

The implementation of the **ScoreSight: EPL Team Points Tally and Top Goal Scorer Prediction System** involves the integration of data analytics, machine learning, and web technologies to create an efficient football prediction platform. The system is developed using Python as the primary programming language, while Streamlit is used to create an interactive web-based application. Additional frontend technologies such as HTML, CSS, and JavaScript are utilized to enhance the user interface and improve user experience.

The implementation process begins with the collection of historical English Premier League (EPL) datasets containing team and player statistics from previous seasons. These datasets include important attributes such as matches played, wins, draws, losses, goals scored, assists, shots on target, goal difference, and total points earned. The collected data is stored in structured formats and prepared for further processing.

The next stage focuses on data preprocessing. The raw datasets often contain missing values, duplicate entries, and irrelevant attributes that can negatively affect model performance. Various preprocessing techniques are applied to clean the data, remove inconsistencies, and transform the information into a machine-learning-compatible format. This stage ensures that the dataset is accurate, reliable, and suitable for predictive analysis.

After preprocessing, feature engineering techniques are implemented to identify and extract the most significant attributes influencing team performance and goal-scoring ability. Relevant features are selected based on their impact on prediction outcomes. These optimized features improve model efficiency and contribute to generating more accurate forecasts.

The machine learning implementation is carried out using the Scikit-learn library. The processed dataset is divided into training and testing datasets, allowing the model to learn from historical football statistics and evaluate its prediction capability. The Random Forest Regression algorithm is employed because of its ability to handle complex relationships between multiple features and provide reliable prediction results. The trained model is capable of estimating future EPL team points tallies and identifying potential top goal scorers.

The prediction model is integrated into a Streamlit application, which acts as the primary interface between the user and the machine learning system. The application allows users to access football analytics, view historical

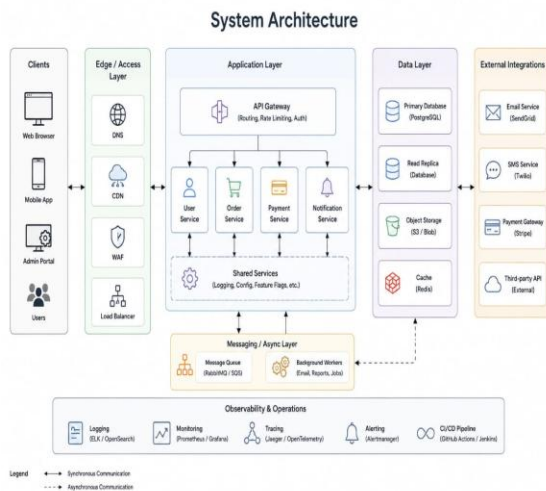


Fig. 1. System Architecture of the Scoresoght-predicting EPL point Tally, Top Scorer.

statistics, and generate predictions through an interactive environment. HTML, CSS, and JavaScript are incorporated to improve page design, navigation, and responsiveness, making the application visually appealing and easy to use.

The implementation also includes result visualization features. Prediction outputs are displayed through tables, charts, and graphical representations that help users understand team rankings, projected points, and player scoring trends. These visualizations provide meaningful insights and make complex analytical information easier to interpret.

Extensive testing is conducted during implementation to ensure system reliability and accuracy. The prediction results are evaluated using historical data and performance metrics. Any identified issues are corrected through model optimization and interface improvements. This testing process helps enhance the overall quality and effectiveness of the system.

The final implementation successfully combines machine learning, football analytics, and web technologies into a unified platform capable of generating EPL team points tally predictions and top goal scorer forecasts. The developed system provides a practical application of data science in sports analytics and demonstrates the effectiveness of machine learning techniques in football performance prediction.

## VII. RESULTS AND DISCUSSION

The results obtained from the **ScoreSight: EPL Team Points Tally and Top Goal Scorer Prediction System** demonstrate the effectiveness of machine learning techniques in football analytics and performance forecasting. The system was trained and tested using historical English Premier League (EPL) datasets containing team and player statistics from previous seasons. These datasets included important attributes such as matches played, wins, draws, losses, goals scored, assists, shots on target, goal difference, and total points. After preprocessing and feature engineering, the data was used to train the Random Forest Regression model for prediction purposes.

The developed model successfully identified patterns and relationships within historical football data, enabling it to generate predictions for EPL team points tallies and potential top goal scorers. The prediction results showed that machine learning algorithms can effectively analyze large volumes of sports data and provide meaningful forecasts based on previous performance trends. The model was able to estimate future outcomes by considering multiple statistical factors simultaneously,

making the prediction process more reliable than traditional manual analysis methods.

The EPL Team Points Tally Prediction module generated projected point totals for different teams based on their historical performance records. The results indicated that teams with strong historical consistency, better goal differences, and higher winning percentages generally achieved higher predicted point totals. The prediction outputs provided valuable insights into potential league standings and team performance trends for upcoming seasons.

Similarly, the Top Goal Scorer Prediction module analyzed player statistics and identified players with a high probability of becoming leading goal scorers. Factors such as goals scored in previous seasons, assists, shots on target, playing time, and scoring consistency significantly influenced prediction outcomes. The generated forecasts highlighted the importance of historical performance data in identifying potential top-performing players.

The Streamlit-based application successfully displayed prediction results through an interactive and user-friendly interface. Users could easily access football statistics, generate predictions, and visualize analytical outcomes using charts, tables, and graphical representations. The visualization features improved result interpretation and allowed users to compare predicted values with historical performance data effectively.

During system evaluation, the prediction model demonstrated satisfactory accuracy and consistency. The use of Random Forest Regression contributed to improved prediction performance due to its ability to handle complex datasets and capture nonlinear relationships among features. The model produced stable results across different testing scenarios and effectively minimized prediction errors through ensemble learning techniques.

The results also revealed certain limitations associated with football prediction systems. Factors such as player injuries, transfers, managerial changes, team strategies, and unexpected match outcomes can influence actual season performance and may not always be fully represented in historical datasets. Despite these challenges, the system generated meaningful predictions that closely reflected historical trends and performance patterns.

Overall, the experimental results confirm that the proposed ScoreSight system is capable of providing reliable football analytics and prediction services. The combination of data preprocessing, feature engineering, machine learning, and interactive visualization creates an effective framework for EPL team points tally forecasting and top goal scorer prediction. The findings demonstrate the practical application of machine learning in sports analytics and highlight its potential for supporting data-driven decision-making in modern football.

## CONCLUSION

The ScoreSight: EPL Team Points Tally and Top Goal Scorer Prediction System successfully demonstrates the application of machine learning techniques in football analytics and sports prediction. The system was developed to analyze historical English Premier League (EPL) data and generate predictions regarding team points tallies and potential top goal scorers. By utilizing data preprocessing, feature engineering, and Random Forest Regression, the system effectively identifies patterns within football statistics and produces meaningful prediction results.

The project highlights the importance of data-driven decision-making in modern sports analysis. Through the use of historical team and player performance records, the developed model provides valuable insights into future league outcomes and individual player achievements. The implementation of machine learning algorithms enables automated analysis, reducing the dependency on manual forecasting methods and improving prediction efficiency.

The integration of Streamlit with HTML, CSS, and JavaScript provides a user-friendly platform where users can interact with the system, explore football statistics, and view prediction results through intuitive visualizations. The generated charts, tables, and analytical reports enhance result interpretation and make the system accessible to football enthusiasts, analysts, researchers, and students.

Experimental results indicate that the proposed system can effectively predict EPL team standings and top goal-scoring performances based on historical data. Although factors such as injuries, transfers, tactical changes, and unexpected match events may affect actual outcomes, the developed model demonstrates satisfactory accuracy and reliability in forecasting football performance trends.

Overall, the ScoreSight system successfully combines machine learning, data analytics, and web technologies to create an efficient football prediction platform. The project contributes to the growing field of sports analytics and demonstrates how historical data can be transformed into valuable predictive insights. The developed system serves as a practical example of applying artificial intelligence and machine learning techniques to solve real-world problems in football analytics and performance prediction.

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